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ORIGINAL PAPER

Seroprevalence of Hepatitis B Infection Among a Population of Blood Donors in the Kpandai District of Northern Ghana: A Five-Year Retrospective Study from 2014 to 2018

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Abstract

Hepatitis B virus (HBV) infection is an infectious disease that is specific to the liver, causing both acute and chronic hepatitis. This study determined the trend of HBV infection among blood donors in the Kpandai District Hospital from January 2014 to December 2018. Retrospective analysis of donor profiles for a period of 5 years was used to assess the distribution of seropositive cases in relation to age, sex, and trend. The data was collected using pre-designed questionnaire and managed using Microsoft Excel spread sheet 2013. The overall sero-prevalence of HBV was 8.2% (230/2802). The year to year analysis depicted significant trends in reduction of HBV infection among the study population at a rate of 13 percentage points above the mean from 2014 to 2015, and 4.3 percentage points below the mean from 2016 to 2018 across the period. The prevalence in males was 8.4% (225/2687) and that of the females was 4.4% (5/115) and the highest rate of infection was among those below 30 years 9.8% (163/1666). Although the general prevalence of HBV infection was high (8.2%), there was a significant trend reduction in rate of infection across the study period. This suggests that the public health interventional strategies put in place to manage the HBV infection in Kpandai District seem to be working.

Keywords Hepatitis B virus · Blood donors · Seropositive · Trend

Introduction

Hepatitis B virus has been described as the most infectious known virus due to its ability to remain infectious outside the human host for not less than 1 week before losing infectivity [1]. Body fluids such as blood, semen, vaginal secretions among others constitute the mediums through which the virus is transmitted (WHO [1]). Beside vertical transmission, studies revealed that blood transfusion is the commonest means of HBV transmission in developing countries [2]. Millions of individuals receive blood or blood products yearly, indicating a high tendency of transfusion transmitted HBV infection among the population, [3]. The World Health Organization reported in 2017 that, out of every 10 units of blood it is likely one is Hepatitis B seropositive [4]. According to [5], blood transfusion is an indispensable component of health care delivery with regards to patient management in that, it is an emergency life-saving measure that is taken to provide support for patients who lost excess blood through accidents, medical, surgical or obstetric conditions [6]. Transmission of Hepatitis B virus via blood transfusion is an outstanding occurrence among others and capable of posing significant long term detrimental side effects including liver cancers [7].

Globally it has been estimated that, over 2 billion people are living with hepatitis B infection out of whom in excess of 350 million are chronically infected [8]. The sub-Saharan African countries and some parts of Asia are known to be the WHO sub-regions with high endemic hepatitis B infections where about 5–10% of the adult population are

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chronically infected [9]. In Ghana the prevalence is estimated to be between 4.8 and 16% [10]. Olawumi et al. [5] stated that, although several efforts including mass vaccinations and early childhood immunizations have been put in place to curb the spread of Hepatitis B infection, the prevalence is still high especially in developing countries, including Ghana.

In Ghana, research has shown that lack of adequate information is a reason for the continuous spread of the disease [11]. The risk of transmission of HBV through unscreened blood for transfusion in the area is not known and few studies have been conducted on the prevalence of hepatitis B infections in Ghana [12, 11]. Consequently, there is scarcity of data on the prevalence of HBV infection in the Kpandai district of northern Ghana. This study was therefore aimed at providing information on the prevalence, trend and associated socio-demographic factors for HBV infection in the district.

Methods

Study Design

This was a hospital based retrospective study using donors' record from the blood bank of the Kpandai District Hospital (KDH) laboratory from January 2014 to December 2018. The hospital is located in Kpandai, the district capital of Kpandai District in the northern region of Ghana. The hospital is a primary level facility and is the principal hospital in the district with a 54 bed capacity. The hospital laboratory employs an immunochromatographic test which is a one-step test to assay for sero-positivity to Hepatitis B surface antigen.

Study Population/Eligibility Criteria

Voluntary and non-voluntary blood donors, who presented at the hospital's blood bank during the study period, were involved in the study. Data from prospective donors between the ages of 16–60 years and passed a history screen of their medical and physical state and pre-donation screening test for hemoglobin level, blood pressure, pulse and weight and were screened for Hepatitis B were considered in the study.

Sampling Method

A total sample size of 2802 archived data on blood donors (both voluntary and non-voluntary donors) was retrieved from the hospital's blood bank using a pre-designed data capturing instrument.

Informed Consent

Due to the fact that this was a retrospective study and getting patients' consent was going to be difficult, we obtained written permission from the Kpandai Distract Hospital making sure that they had obtained the consent of the patients in the past for using their data in the future. Donor information has also been kept confidential.

Power and Statistical Analysis

The blood transfusion service clinical record sheets of the laboratory and screening registers were reviewed and relevant data for the study collated onto a pre-designed data capturing questionnaire. Entries from the questionnaires were double-entered into a predesigned electronic database using Microsoft Excel 2013, verified and cleaned. Finally, the verified and cleaned data was exported to IBM Statistical Package for the Social Sciences (SPSS Inc., Chicago, USA) version 22.00 and analysed by doing cross tabulation to generate the proportions. Differences between proportions and trend analysis were done using Fisher's exact test and chi square test for trends where appropriate. Graph pad prism was also employed to generate some of the graphs. A p-value of < 0.05 was considered significant in all statistical comparisons.

Results

During the 5-year study period, a total of 2802 prospective donors constituted the population of donors who were screened for HBsAg. The donor population comprised people from the Kpandai district as well as people coming from the neighbouring districts. Majority (85.6%) were resident in the district with 73.1% of them in rural communities. Out of the total number of donors, 2687 representing 95.9% were males and 115 (4.1%) were females. In terms of the donor age categories, those within 20–29 years were 43.25% while those between 30–39 years were 34.83%. The rest were 9.21%, 8.96%, and 3.75% for the 40–49 years, < 20 years, and ≥ 50 years respectively (Table 1). A total of 97.79% of the donors were non-voluntary donors whereas the remaining 2.21% were voluntary donors.

In this study, 230 donors were found to be positive for HBsAg with an overall seroprevalence of 8.21%. Of the 2740 non-voluntary donors, 8.39% were seropositive while there was no positive case among the voluntary donors. For the community levels, 8.88% (183/2049) of rural residents were positive as against 6.37% (48/753) of those residing in the urban communities. Furthermore, 8.25% (198/2400)

Table 1 Demographic characteristics of blood donors at the Kpandai District Hospital

Parameter	Frequency	Percentages (%)
Total	2802	100
Gender		
Male	2687	95.9
Female	115	4.1
Age category		
< 20	251	8.96
20–29	1212	43.25
30–39	976	34.83
40–49	258	9.21
≥ 50	105	3.75
Community level		
Rural	2049	73.13
Urban	753	26.87
Residential status		
Resident	2400	85.65
Non-resident	402	14.35

Data is presented as frequency and corresponding percentages

and 7.96% (32/402) of residential and non-residential donors were found to be seropositive respectively. There were significant differences in the infection rate among the donor type (p = 0.0318) and the community level of donors (p = 0.0354) but no significant difference in infection rate among donors with regards to their residential status (p = 0.9221) (Table 2).

Majority of the donors were found to be blood group 'O' (61.6%) followed by blood group 'B' (24.70%). Similarly blood group A, and AB were 13.06% and 0.64% respectively. With regards to the Rhesus (RH) factor, 88.8% was Rh-D positive while the remaining 11.2% was negative (Table 2). Analysis of the distribution of HBV infection among the ABO blood group and Rh-D status of donors is presented in Table 3. The highest rate of infection recorded was among blood group A (10.9%) followed by 9.97% for those in group 'B' and 6.95% among blood group O donors. The lowest rate of the HBV infection was among blood group AB (5.56%). Those who were Rh-D positive showed a higher rate of infection (8.16%) as compared to those with Rh-D negative genotype (7.67%) (Table 3).

The year to year analysis showed a significant trend in reduction of Hepatitis B virus infection among the study population across the period (Fig. 1). The highest rate of infections among participants was 15.02% in 2014. This was followed by 10.9% in 2015, 6.13% in 2017, 4.55% in 2018 and 2.22% in 2016. Thus, the average rates of infections trended significantly by 13% percentage points above the mean in 2014 and 2015, and 4.3% below the mean between 2016 and 2018 (Fig. 1).

Furthermore, the rate of HBV infection among the male population was 8.4%, and the female was 4.4% across the study period. Specifically, by the year to year analysis, the highest rate of 15.3% recorded among the male population was in 2014. This was followed by 11.0% in 2015 and the lowest rate of infection among the males across the period was in 2016 (2.34%). However, the highest rate of 8.7% recorded among the female population was in 2015, followed by 6.1% among the females in 2017 and there was no prevalence among the females in both 2016 and 2018. The variations in proportion of HBV infection by the gender distribution across the period were not statistically significant, with infection projections tilted towards the male donor population (Fig. 2).

The age range of the blood donors was from 16 to 60 years with mean age of 29.94 years and a standard

Table 2 Prevalence of HBV infection by donor type, residential status and community level of the study population

Parameter	Frequency (%)	HBsAg positive (%)	p-value
Total	2802 (100)	230 (8.21)	–
Donor type			
Non-voluntary	2740 (97.79)	230 (8.39)	0.0318
Voluntary	62 (2.21)	0 (0.00)	
Residential status			
Resident	2400 (85.65)	198 (8.25)	0.9221
Non-resident	402 (14.35)	32 (7.96)	
Community level			
Rural	2049 (73.13)	183 (8.88)	0.0354
Urban	753 (26.87)	48 (6.37)	

Data is presented as frequency and corresponding percentages in parenthesis

p value for trends is significant at 0.05

Table 3 The distribution of HBV infection among the ABO blood group and Rh-D status of donors

Parameter	Total donors (%)	HBsAg		*p*-value
		Positive (%)	Negative (%)	
ABO				
A	366 (13.06)	40 (10.93)	326 (89.07)	0.0167
B	692 (24.70)	69 (9.97)	623 (90.03)	
O	1726 (61.60)	120 (6.95)	1606 (93.05)	
AB	18 (0.64)	1 (5.56)	17 (94.44)	
Rh-D				
Positive	2489 (88.83)	203 (8.16)	2286 (91.84)	0.8505
Negative	313 (11.17)	24 (7.67)	289 (92.33)	

Data is presented as frequency and corresponding percentage in parenthesis

p for trends is significant at 0.05

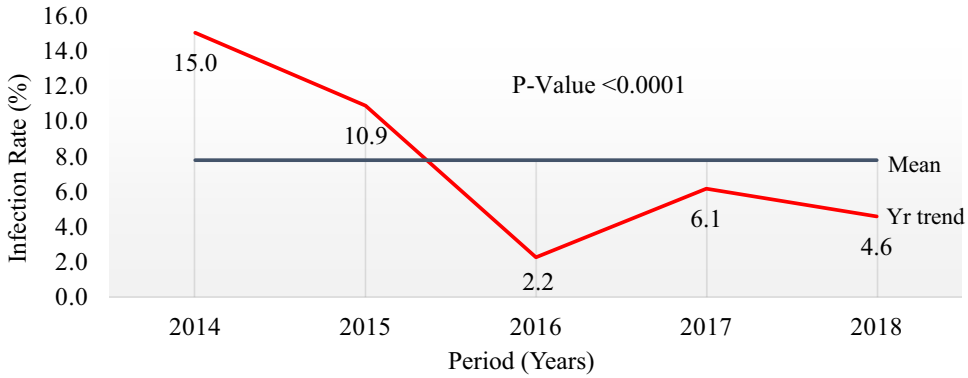


Fig. 1 Trend of HBV infection across study period from 2014 to 2018

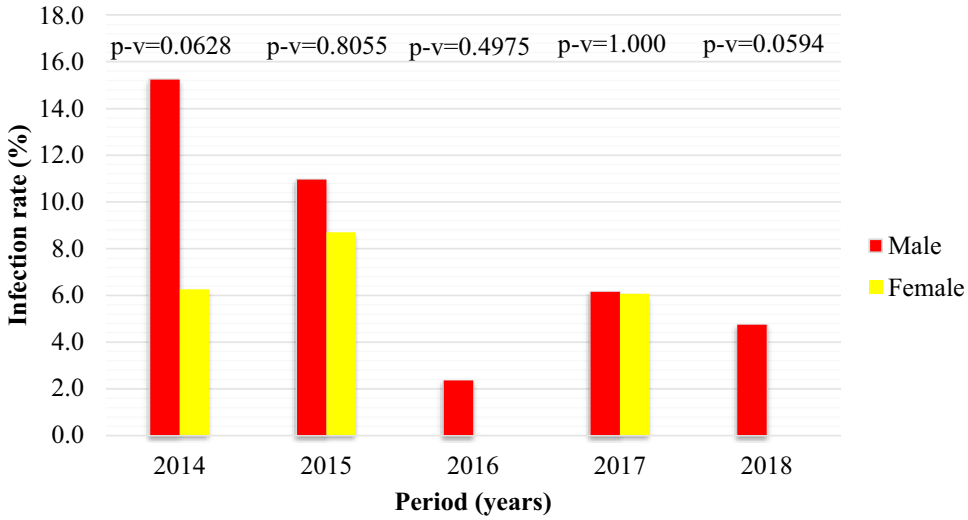


Fig. 2 Variations in proportion of HBV infection by the gender distribution across the period

deviation of 8.62 years. For male blood donors, their age ranged from 16 to 60 years with mean age of 30.09 years and a standard deviation of 8.63 years. The age range for the female donors however, was from 16 to 58 years with a mean and standard deviation of 26.27 and 7.70 years respectively. For HBsAg positive donors, the age ranged from 16 to 55 years. The mean age was 28.28 years with a standard deviation of 7.01 years. For the males infected with HBV, the age range was from 16 to 55 years and a mean age and standard deviation of 28.36 and 6.99 years respectively. Meanwhile, the age range of infected females was from 17 to 35 years, mean of 24.60 years and a standard deviation of 7.23 (Table 4).

Figure 3 shows the stratification of the year-on-year trend and age-related distribution of HBsAg positive participants. Generally, there was a significant declining trend of HBV infection as the age increased across the age categories ($p < 0.0001$). The highest number of infections was among participants of 20–29 year age group. This was followed by those within 30–39 years and the least was among those above 50 years of age. This was a common observation among all the years. The year 2014 however, showed the peak of incidence by recording 45 HBsAg positive cases among the 20–29 year age group of donors representing 17.24% in that year. This was followed by 2015 with 35 cases, representing 13.36%, and then 2017 with 17 cases at 6.64% prevalence. The years 2018 and 2016 had 12 (5.45%) and 6 (2.82%) positive cases respectively. The lowest prevalence was recorded among the ≥ 50 years age group in each year across the period, with 2014 alone not recording a zero prevalence as two out of the eight ≥ 50 years participants in

Table 4 Univariate analysis of host-gender interaction with transfusion transmissible Hepatitis B infection using age as a continuous variable of the statistical population screened

Population	Age range	Mean	SD	p-value
Blood donors	16–60	29.94	8.62	< 0.0001
Male	16–60	30.09	8.63	
Female	16–58	26.27	7.7	
HBsAg infection	16–55	28.28	7.01	< 0.0001
Male	16–55	28.36	6.99	
Female	17–35	24.6	7.23	

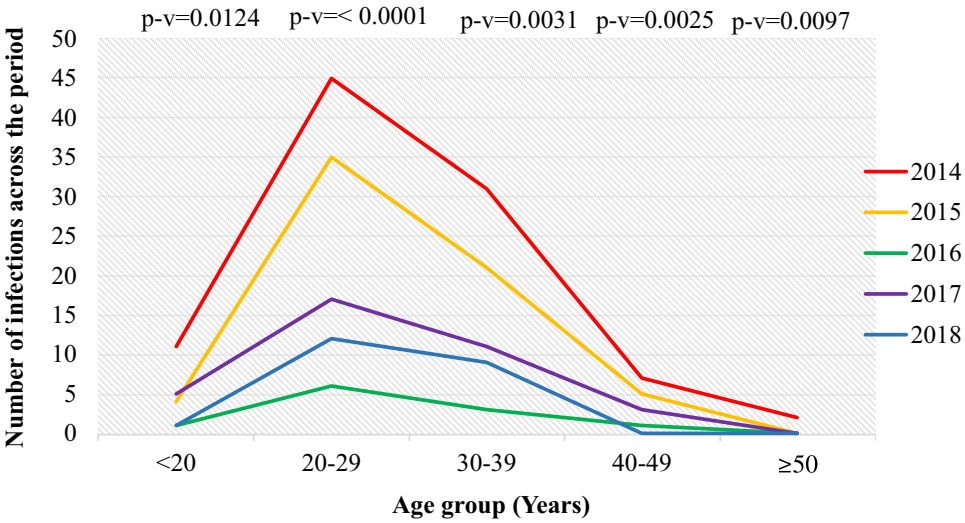


Fig. 3 Stratification of the year-on-year trend and age-related distribution of HBsAg positive participants

that year were sero-positive for the HBsAg. Nonetheless, the year 2018 also recorded a zero prevalence among the 40–49 years age group. The variation among the years for each age category showed statistical significances (< 20; p = 0.0124 20–29; p < 0.0001, 30–39; p = 0.0031, 40–49; p = 0.0025, ≥ 50; p = 0.0097).

Figure 4 is an illustration of the age trend stratified by gender. Generally, the prevalence declined as the age increases among both gender categories but without statistical significance. The highest infection rate among the male was recorded among the 20–29 year group 9.8% (113/1158). The lowest prevalence for the males was 1.92% which was recorded among ≥ 50 years age group following a steady decline of prevalence from 9.76 to 7.78%, to 6.37% amongst the 20–29, 30–39 and the 40–49 year groups respectively. Unlike in the case of their male counterparts, the adolescents (< 20 years) had the highest prevalence of 7.14% (2/28) among the females. The 30–39 year group experienced the second most prevailing HBV infection rate of 4.0% among the females as the 20–29 year group emerged third with a prevalence of 3.7%. Both 40–49 and ≥ 50 years age groups had the lowest prevalence among the females with each of them recording zero positive cases. For the gender variation among the various age categories, only 40–49 years age group showed a significant variation between males and females (p = 0.0289), the rest were statistically stable (Fig. 4).

Discussion

This study sought to find prevalence and trends of HBV infection among a population of blood donors in the Kpandai District of Northern Ghana. In this study, 2802 blood donors visited the Kpandai District Hospital across the period of five years, from January 2014 to December 2018. The study uncovered dominance of non-voluntary donors, male donors as well as donors younger than or equal to 30 years of age. These findings are consistent with a previous study conducted in Tamale, Ghana Dongdem et al. [2]. The findings are also in line with earlier studies conducted in the Ashanti Region Nkrumah et al. [12], and Volta Region Osei et al.

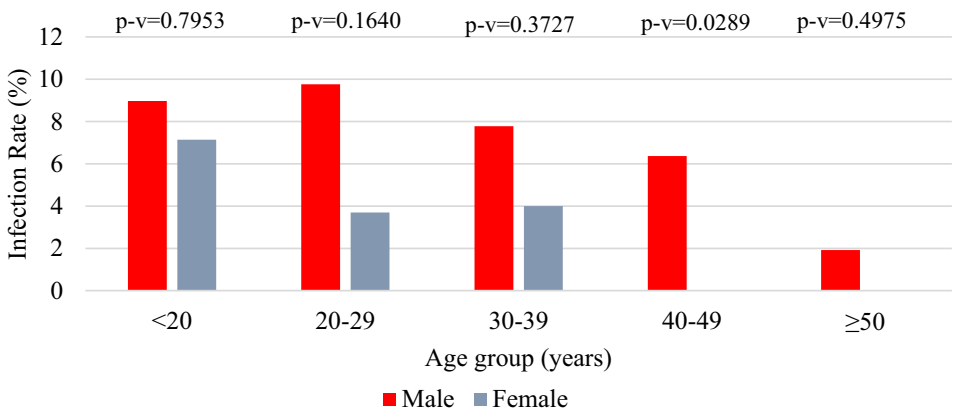


Fig. 4 Age and trend of HBV infection stratified by gender

[13] of Ghana. Similar findings were also reported in Nigeria Okwesili et al. [14] and Tirupati-India Prasad et al. [15]. Though concrete explanations to this regular occurrence of male dominance in the donor population among Ghanaians are not fully established, there are extrapolations that, certain beliefs, socio-cultural practices as well as male proactiveness in taking decisions over females could be some of the influences for this recurrent phenomenon [16, 2, 17]. Other famous reasons explain that menstruation and child bearing do contribute significantly to lesser female donors than males as individuals in these said categories are usually not allowed or otherwise deferred from donating blood. Others too hold the misconception that blood donation is just an activity for males and not females [8]. Lack of education on the need for people to donate blood voluntarily could be the reason for the low turnout of voluntary donors while vibrancy could be the major explanation for the dominant younger donors in this study.

In all, 230 donors were found to be positive for HBsAg with sero-positivity rate of 8.21%, indicating a high prevalence in accordance with WHO guidelines (WHO [1]). The overall prevalence of 8.21% found in the current study is consistent with the prevalence rates of previous studies by Walana et al. [17] and Osei et al. [13]. Other studies conducted earlier by Nkrumah et al. [12] and [18] recorded relatively higher prevalence of 13.8% and 13.3% in 2011 and 2014 respectively. In a meta-analysis by Ofori-Asenso and Agyemang, a 12.3% nationwide prevalence was recorded while the Northern regional prevalence was slightly higher than the national average with an estimated prevalence of 13.1% [11]. This study is also in consonance with previous studies in Nigeria Fasola et al. [19] and Ethiopia [20] where prevalence of 8.5%, and 10.9% respectively were recorded. The outcomes of the current study is in synchrony with others cited above and therefore goes to confirm the classification of Ghana and the Sub-Saharan African countries as a HBV endemic region [19, 21, 20]. India [15] and Italy [22] lower prevalence of 1.76% and 0.33% were recorded among blood donors respectively. Although the prevalence of the current study is relatively lower compared to what had been recorded earlier in other parts of Northern region, it still indicates that HBV infection in Ghana is within the high endemic category since it is $\geq 8.0\%$ [2, 23]. The high HBV prevalence in the Kpandai District and previously recorded high prevalence in other parts of the country among blood donors could be attributed to the higher percentage of non-voluntary donors who are a high risk group than voluntary non-remunerated donors, lack of adequate awareness and inability of the inhabitants to afford the cost involve to take the three shots of hepatitis B vaccination.

Despite the high prevalence, there was an appreciable decreasing trend of HBV infection across the study period after a peak incidence was observed in the first year [2014 (15.02%)], ($p < 0.0001$). This outcome was not different from the findings of Ofori-Asenso et al. in their meta-analysis in Ghana [11], and Prasad et al. [15], in their study conducted in India from 2012 to 2017. The decreasing trend in HBV prevalence among the population in the district may reflect the effect of the periodic screening and vaccination programmes that are being organized by non-governmental organizations (NGOs) and other cooperate organizations, like churches in the district. The change can also be partially attributed to the increasing awareness of HBV infection in the district as done by healthcare professionals who embark on educational campaigns on the disease, and go round communities in the district for screening and vaccination against the HBV.

Comparable to the pattern known globally, males were the dominant category in terms of the donor population as well as the positive cases for HBV infection. The seroprevalence of HBV was at 8.37% among the male donors as compared to the intermediate 4.35% among their female counterparts in conformity with previous findings (Dongdem et al. [2], Walana et al. [17], Yadav et al. [24] and Arya et al. [25]) where males also dominated in terms of donor population and prevalence of HBV infection. In contrast with this study, other studies recorded high prevalence of HBV among female donors compared to the males Nkrumah et al. [12], Lokpo et al. [26, 27] and Osei et al. [13].

The declining trend also was more consistent among the male donors but varied across the years among the female donors as seen in, Fig. 2. However, this observed difference among the gender category was not statistically significant ($p = 0.1235$). Nonetheless, in an experiment with laboratory mice and further elution of the abnormal form of this protein from the serum of HBV infected males, a group of Chinese scientists concluded that, the variant form of Apo lipoprotein A-1 produced only in the liver of males was responsible for the HBV preference for males over females [28]. Moreover, in a survey it has been shown that more males have multiple sexual partners than females [29], which could explain the high HBV infection among males than females. The high prevalence of HBV among males may also just be attributable to the excessive dominant male over female donors leading to more males being the positives than females.

This study revealed that, majority of seropositive donors were young (20–30 years) similar to what had been reported earlier Nkrumah et al. [12], Dongdem et al. [2], Walana et al. [17]. In contrast, Osei et al. [13] reported that those within 30–40 years had the highest prevalence of HBV among blood donors ([13] with the peak cases observed among people between 30 and 40 years of ages. Despite the variations in this study and that of Osei and colleagues, people within the age range of 20–40 years have been classified as the most sexually and economically active group [30] and are more likely to have multiple sexual partners and extreme sexual

activity which further put them at risk of getting HBV infection. Perhaps as well, since the younger age category form majority of the donor population, there is a greater chance of obtaining a higher proportion of positive cases among them than the older ones. Again, since it is the same younger ones who will eventually transit into adult donors, with a higher number of them being infected at the younger age, first such people in the future cannot donate again thereby could lead to reduction in the number of positive cases among donors at their adult age. Secondly, some might have also died of the infection before or during advancement in age due to their weakened immunity and increasing mortality rate of the HBV, hence reducing the number of cases in the adult population at their adult age.

Furthermore, the present study shows that prevalence of HBV is higher amongst non-voluntary donors (8.39%) and the same was found in other similar studies carried out by Okwesili et al. [14] and Prasad et al. [15]. The outcome may be suggesting that voluntary donors are certain and are aware of their health status. This is because, before they decided to do a voluntary donation, they might have received education on donation as such are able to self-exclude themselves leading low positivity detection among them. On the other hand, because the non-voluntary donors are doing the donation on a condition or out of compulsion by circumstance, they may be unaware of their health status or lack knowledge about blood donation. This also suggests that the voluntary donors may not be a true representation the natural population as most of them come to do donate already well informed about their health status unlike the non-voluntary donors who may be quite naïve about theirs and hence may represent the natural population more accurately. This problem can be controlled by careful elimination of commercial, and replacement donors and encouragement of voluntary donations.

From the study, 8.88% (183/2049) of the rural resident donors were seropositive for HBsAg compared to 6.37% (48/753) of those dwelling in urban communities. There was a statistical significance for these variations ($p = 0.0354$). The results agree with the findings of [18] and [11] in their studies in 2014 and 2016 respectively. This might be as a result of lack of adequate information and education campaign about the infection in the rural areas leading to the uncontrolled spread of the virus hence the high prevalence. Additionally, the people living in the rural areas may share common and basic items like toothbrush, sponges and towels which have the potential to and could facilitate the spread of the virus among them easily. More targeted educational campaigns for awareness, increase in literacy and vaccination in the rural areas will be very useful in addressing this situation.

In this study, blood group A individuals were the most infected as compared to 'AB' persons being the least. There was no significant difference between the Rh-D blood group categories though Rh-D positive individuals were about 90% of those who tested positive for HBsAg. It has been suggested that both blood group 'O' and Rh-D positive persons are more susceptible to HBV infection as compared to the other types of ABO and Rh-D negative [31] which is contrary to this current study. In another study it was found that groups B and AB as well as Rh-D negative were more associated with the infection than the others [32]. The discrepancies in previous studies and our study need further exploration as such researchers have still not fully understood the association between HBV infection and the blood group types of individuals as different studies have reported different associations.

Conclusion

The overall prevalence of HBV infection in the study population was 8.2% with significant decline in sero-prevalence of the infection among blood donors across the study period, from 2014 to 2018. This shows that the public health interventions in place at the District are making progress. However, significantly higher prevalence was recorded in non-voluntary blood donors compared to voluntary blood donors as well as in donors aged between 20 and 30 years as compared to the other age group categories.

We therefore recommend rigorous immunization backed by effective health education which should specifically target the high risk groups. Furthermore, the use of advanced screening techniques should be encouraged in order to effectively help in the prevention and to sustain the declining trend of this infectious disease and also to serve as foundation pillars for safe blood transfusion in the community in particular and the country as a whole.

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Compliance with Ethical Standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethical Approval The protocol used for the study was approved by the University of Health and Allied Sciences' Research Ethical Committee (Protocol Number: UHAS-REC/A.4 [227] 18-19).

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